Report CSS

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Introduction

This research is about CSS, one of the most important languages in web design. CSS stands for Cascading Style Sheets, and it controls the look of a website. When we build a page with HTML, it looks plain, just text and images. Here comes the role of CSS. It makes the page beautiful by adding colors, fonts, and organizing the layout.

The nice thing about CSS is that it separates content from design. So we write the style once and use it everywhere, which makes the code cleaner and easier to edit.

In this research, I will talk about how to change colors and fonts, and how to design a website layout using CSS.

Ways to Add CSS

There are three ways to add CSS to a page:

1. External CSS: We write the code in a separate file with .css extension and link it to the HTML page using <link>. This is the best way because one file can be linked to many pages.

2. Internal CSS: We write the code inside a <style> tag in the <head> of the same page. We use this when we have only one page.

3. Inline CSS: We write the style directly inside the HTML tag using the style attribute. It is fast but makes the code messy.

CHANGING COLORS WITH CSS

- By name: like red, blue, green.
- Hexadecimal: like #FF0000, and this is the most used one.
- RGB: like rgb(255, 0, 0).
- RGBA: like rgba(255, 0, 0, 0.5) and the last number is for transparency.

Main Properties

- color: changes the text color.
- background-color: changes the background color.
- border-color: changes the border color.

```css
body {
  background-color: #f4f4f4;
  color: #333333;
}

h1 {
  color: #1e90ff;
}
```

CONTROLLING FONTS

- font-family: to choose the font type. We write more than one as a backup, like:
- font-family: "Cairo", Arial, sans-serif;
- font-size: to change the font size.
- font-weight: to make the font bold.
- font-style: to make the font italic.
- text-align: to move the text right, left, or center.
- line-height: to control the space between lines.

```
p {
  font-family: "Cairo", Arial, sans-serif;
  font-size: 16px;
  line-height: 1.8;
}
```

Website Design and Layout

CSS also controls the whole layout of the page. The most important techniques are:

1. The Box Model: Every element is like a box that has content, padding, border, and margin. Understanding this helps us control spaces easily.
2. Flexbox: Used to arrange elements in a row or column. It is useful for navigation bars and cards.
3. Grid: Divides the page into rows and columns. It is good for big layouts.
4. Responsive Design: The website should look good on all screens like phones and computers. We use @media to change the design when the screen size changes.
5. Effects: CSS gives us transition for smooth changes and animation for moving effects.

```css
.container {
  display: flex;
  gap: 20px;
}

@media (max-width: 768px) {
  .container {
    flex-direction: column;
  }
}
```

References

1. MDN Web Docs – CSS Section.
2. W3Schools – CSS Tutorial.
3. "HTML and CSS: Design and Build Websites" – Jon Duckett.

Thank **You**